Hydraulic Bottle Jack: Disassembly, Measurement, and Analysis

Student Handout – Mechanical and Hydraulic Systems

Name ____________________________ Date __________________

1 Introduction

A bottle jack is the smallest complete hydraulic power system you will ever hold. It contains a reservoir, a positive-displacement pump, two check valves, a directional control element, an actuator, and a set of passages, and it performs the full cycle of a hydraulic system: draw fluid, pressurize it, do work with it, hold the load, and return the fluid to tank. Everything you will later study in an industrial circuit is present here, as Figure 1 shows, at a scale you can hold in one hand and take apart in a single laboratory period. The equations you use on this jack are the same equations used on a 200 ton press.

> **The jacks in this laboratory are drained.** There is no oil in them and no load will be lifted. Every result you produce is therefore a **prediction** derived from geometry, which is itself a legitimate engineering activity: you are determining the performance of a machine from its dimensions alone. It also means that nothing here validates the ideal model against a real machine, so report every result as a prediction rather than as a measurement.

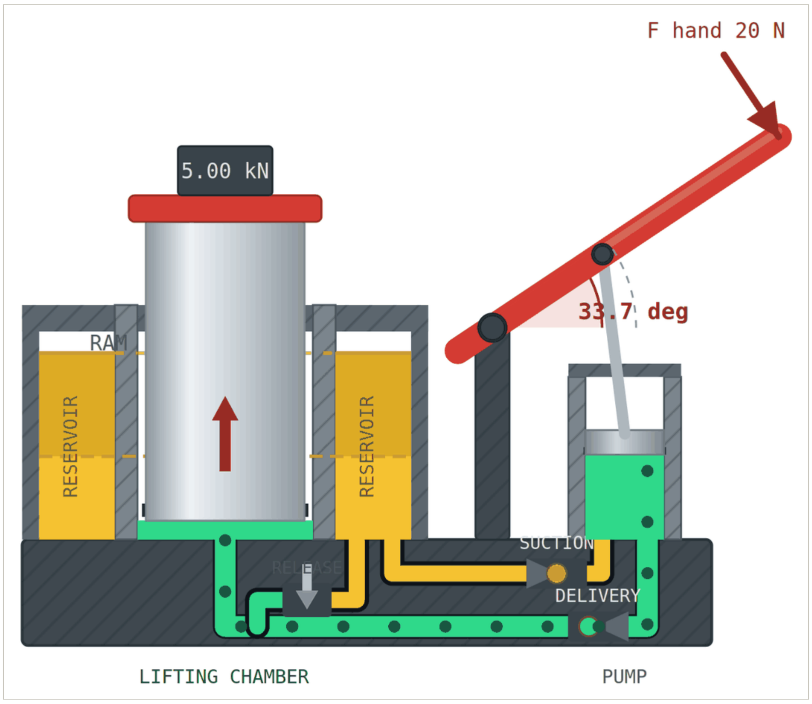

Figure 1. Cross section of a bottle jack during the pressure stroke, drawn by the companion simulation. Amber is reservoir oil at atmospheric pressure and green is oil at lifting pressure, so the boundary between the two colors is the delivery check ball itself. The plunger is descending, the suction ball is seated, and oil is passing along the base gallery into the chamber beneath the ram. The two amber columns are one annular volume surrounding the lifting cylinder, seen twice in section. Dimensions are omitted here because measuring them is your task; the simulation displays them when the dimension overlay is switched on.

1.1 Companion simulation

An interactive cross-section of the jack accompanies this laboratory. It shows oil moving between the reservoir, the pump chamber, and the lifting chamber while the handle is operated, with the check valves opening and closing according to the pressure across them. Every dimension in the drawing can be edited, in either SI or US customary units, and the ram rises only by the volume the pump has actually delivered, so the animation obeys conservation of volume rather than approximating it.

Use it before the laboratory to predict how many strokes a full extension requires, and afterwards to check your calculations against your own measured geometry. Note that the simulation and your hand calculations share the same ideal model, so agreement between them confirms your arithmetic and nothing about the physical jack.

> **Simulation.** `https://alibulentkoc.github.io/hydraulic-bottle-jack/simulator/hydraulic-bottle-jack.html`
> Runs in any browser, with no installation and no network connection once loaded.
> Scan the code or type the address; either reaches the same page.
>
> Scan for the simulation

2 Objectives

On completing this laboratory you will be able to:

1. Identify the functional components of a hydraulic bottle jack and trace the fluid path from reservoir to lifting chamber and back.
2. Select and use the correct instrument for each feature: dial calipers, telescoping gauge, small-hole gauge, micrometer, dial indicator, and angle gauge.
3. Apply conservation of volume to predict the lift produced by one full pump stroke.
4. Apply Pascal's principle to relate load, piston area, and system pressure.
5. Analyze the pump handle as a lever and compute the torque and hand force required at rated load.
6. Compute hydraulic power and linear power, and compare operator input to lifting output.
7. Explain, with numbers, why force multiplication in any machine is paid for with distance.

3 Core Principles

The following eight ideas are the reason this device merits a laboratory period. Every question you answer later is one of them with numbers attached.

1. **Pressure is created by the load, not by the pump.** The pump moves fluid. The resistance the fluid meets is what raises pressure. A jack with no load on it develops almost no pressure no matter how hard you pump.

$$\mathrm{P} = \mathrm{F} \,/\, \mathrm{A} \tag{1}$$

2. **Pascal's principle makes the area ratio a force multiplier.** When the delivery check valve is open, the pump chamber and the lifting chamber are one connected body of fluid at one pressure. The same pressure on a larger piston produces a larger force.

$$\mathrm{F\_ram} \,/\, \mathrm{F\_pump} = \mathrm{A\_ram} \,/\, \mathrm{A\_pump} \tag{2}$$

3. **Conservation of volume makes the same area ratio a distance divider.** Oil is treated as incompressible, so the volume that leaves the pump appears under the ram. The larger the ram, the less it rises for that volume.

$$A\_pump * s\_pump = A\_ram * s\_ram \tag{3}$$

4. **Force multiplication is paid for with distance, so power is not created.** Principles 2 and 3 use the same ratio in opposite directions. Multiply force by 250 and you divide speed by 250. Ideal power in equals ideal power out; a real machine returns less.

$$\text{Power} = F * v = P * Q \tag{4}$$

5. **The two multiplications are in series and simply multiply.** The handle is a lever with its own ratio, ahead of the hydraulics. The total advantage of the jack is the product of the two, which is how a 4 lb hand force holds a 1000 lb load.

$$MA\_total = (A\_ram / A\_pump) * (L\_handle / L\_pivot\text{-}to\text{-}link) \tag{5}$$

6. **Two check valves turn reciprocating motion into one-way flow.** The plunger moves back and forth. The valves make the fluid move in one direction only: the inlet ball admits on the up stroke and seals on the down stroke, the delivery ball does the reverse. This is the defining trick of every positive-displacement pump.

7. **A closed valve holds the load with zero input power.** Once the delivery ball seats, the trapped fluid column carries the load indefinitely with no energy supplied. An electric motor holding the same load consumes power continuously. This static holding ability is a principal reason hydraulics is chosen for lifting.

8. **Lowering is metered, not powered.** The release valve does not push the ram down. The load does. The valve controls the rate at which fluid is allowed back to the reservoir, which is throttling, or flow control, in its simplest form.

4 Safety

READ BEFORE BEGINNING

- Wear eye protection during the entire disassembly. Check balls, springs, and seals can leave a bore under spring tension.
- The jacks are drained but not dry. Expect residual oil. Wear nitrile gloves, work over absorbent pads, and wash before leaving.
- Use a magnetic parts tray. Check balls are small, they roll, and a jack is scrap without them.
- Do not apply a load to any jack in this lab. No jack in this room contains fluid, and a jack that appears to hold will not.
- Report any tool that is dropped or any gauge that is damaged. A sprung telescoping gauge produces confident wrong numbers, which is worse than an obviously broken one.

5 Equipment

Per group: bottle jack, dial calipers, telescoping gauge set, small-hole (split-ball) gauge set, 0 to 1 in micrometer, dial indicator with magnetic base, digital angle gauge or protractor, steel rule, hex and socket wrenches, magnetic parts tray, absorbent pads, nitrile gloves, camera or phone for the component photograph.

> **On instrument choice.** A telescoping gauge and a small-hole gauge do not read themselves: you transfer the setting to a micrometer. Reading either one with dial calipers is the single largest source of error in this lab, and because area depends on diameter squared, that error arrives in your answer at double size. The pump bore is near 0.3 in, below the range of most telescoping gauge sets, so it requires the small-hole gauge.

6 Procedure

The order below is not the order of the questions. Measure in this order, because two of the measurements are only available while the jack is assembled and three of them are only available while it is apart.

6.1 Part A: Familiarization, jack assembled

1. Record the manufacturer, model, and the **rated capacity from the nameplate** in Table 1. Convert tons to pounds and record both.
2. Operate the handle through several full strokes and open and close the release valve. Note by feel and sound where the plunger reaches each end of its travel and where the release valve seats.
3. Sketch the jack in your notebook and mark where you expect to find the reservoir, the pump bore, the two check valves, the release valve, and the lifting chamber. You will check this sketch against the opened jack in Part C.

6.2 Part B: Kinematic measurements, jack still assembled

> These two measurements are lost the moment you take the jack apart. Do them first and have a second group member verify each reading before you move on.

1. Mount the **dial indicator** against the top of the pump plunger. Sweep the handle slowly from its lower stop to its upper stop and record total plunger travel. This is the pump stroke s_p. Take three readings and record all three in Table 2.
2. With the **angle gauge** on the handle, record the handle angle at the lower stop and at the upper stop. The difference is the swept angle θ. Record it on the same stroke you measure, not separately.
3. Measure the perpendicular distance from the handle pivot to the point where the linkage connects to the pump rod a, and from the pivot to the center of the operator's grip b. State where on the grip you measured to; this choice changes your answer to question 7 and must be reported.

6.3 Part C: Disassembly and internal measurements

1. Disassemble in a documented order. Photograph each stage and lay parts on the tray in the sequence they came out.
2. Compare the opened jack with your Part A sketch. Correct the sketch and mark the actual flow path: reservoir to inlet check valve to pump chamber to delivery check valve to lifting chamber, and lifting chamber to release valve back to reservoir.
3. Measure the **lifting cylinder bore** D_r with the telescoping gauge and micrometer. Take three readings at different depths and two clock positions. A worn jack is oval and tapered, and the spread between your readings is data about the condition of that jack.
4. Measure the **pump cylinder bore** d_p with the small-hole gauge and micrometer. Three readings.
5. Measure the **reservoir annulus**: inside diameter D_i, which is the outside of the lifting cylinder, and outside diameter D_o, which is the inside of the jack body. Also measure the **height of the annular oil space** H, from the floor of the reservoir to the fill plug or to the top of the annulus, whichever limits the oil. **All three are available only now.** You cannot return for them after reassembly.
6. Measure the **full extension** of the lifting cylinder L, from the seated position to the mechanical stop.
7. Inspect and record the condition of the ram seal, the check balls and seats, and the release valve seat. Note scoring, pitting, or debris.

6.4 Part D: Reassembly

1. Reassemble in reverse order using your photographs. Confirm both check balls are seated in their bores before closing the base.
2. Operate the handle and the release valve to confirm free movement. Return the jack, tools, and gauges to the positions shown on the bench card.

6.5 Part E: Analysis

Complete questions 2 through 13 in Section 8. Carry at least four significant figures through intermediate steps and round only at the final answer. Show the formula, the substitution with units, and the result for every question. An answer with no units is not an answer.

7 Data

Table 1. Identification.

Manufacturer and model		Rated capacity (ton)	
Serial or lab unit number		Rated load W (lb)	

Table 2. Kinematic measurements, jack assembled.

Quantity	Symbol	Reading 1	Reading 2	Reading 3	Mean	Spread
Pump plunger stroke (in)	s_p					
Handle swept angle (deg)	θ					
Pivot to pump rod end (in)	a					
Pivot to operator hand (in)	b					

Note the point on the grip that b was measured to:

Table 3. Internal measurements, jack disassembled.

Quantity	Symbol	Reading 1	Reading 2	Reading 3	Mean	Spread
Lifting cylinder bore (in)	D_r					
Pump cylinder bore (in)	d_p					
Reservoir inside dia (in)	D_i					
Reservoir outside dia (in)	D_o					
Reservoir oil column height (in)	H					
Lifting cylinder extension (in)	L					

Condition notes: seals, check balls, seats, bore scoring

8 Calculations

Use the mean of your three readings for every dimension. Units are US customary throughout: inches, pounds, psi, horsepower, with 1 hp = 550 ft-lb/s = 6600 in-lb/s.

2. Volume displaced per pump stroke

Use the pump bore and stroke from Tables 2 and 3 to calculate the volume of fluid displaced per one full stroke of the piston pump.

```
A_p = (pi/4) * d_p^2
V_p = A_p * s_p
```

A__p = ________________ in^2 V__p = ________________ in^3

3. Lifting cylinder area and lift per stroke

Calculate the cross-sectional area of the lifting cylinder, then use it with your answer to question 2 to determine the lift produced by one full pump stroke.

```
A_r = (pi/4) * D_r^2
V_p = A_r * h    ->    h = V_p / A_r
```

A__r = ________________ in^2 h = ________________ in per stroke

4. Pressure required in the lifting cylinder

Use the rated load from the nameplate and your measurements to determine the fluid pressure required to lift that load.

```
P = W / A_r
```

P = ________________ lb/in^2

5. Downward force on the pump piston

Determine the downward force required on the pump piston to produce that pressure. State which core principle allows you to use the same pressure on both pistons.

```
F_p = P * A_p
```

$F\_p =$ ______________ lb

6. Torque required at the handle pivot

Using the pump force and the pivot to rod end distance, calculate the torque that must be applied at the handle pivot.

```
T = F_p * a
```

$T =$ ______________ lb-in

> **Terminology.** The quantity here is a torque, also called a moment. It is not a moment of inertia, which carries units of lb-in-s^2 and describes resistance to angular acceleration.

7. Tangential force at the operator's hand

Use the pivot to hand distance to determine the force the operator must apply perpendicular to the handle to produce the required torque.

```
T = F_hand * b    ->    F_hand = T / b
check:  F_hand = F_p * (a / b)
```

$F\_hand =$ ______________ lb

8. Hydraulic power at one cycle per second

If you pump at one full cycle per second while lifting the rated load, how much hydraulic power are you generating?

```
Q = V_p * n
Power = P * Q          [ (lb/in^2)(in^3/s) = in-lb/s ]
1 hp = 6600 in-lb/s
```

Q = ________________ in^3/s Power = ________________ hp

9. Time to full extension

Determine how many strokes and how long it takes to extend the lifting cylinder fully at that rate.

```
N = L / h          t = N / n
```

N = ________________ strokes t = ________________ s

10. Extension speed and output linear power

Calculate the speed of extension, then use it with the rated load to find the linear power delivered to the load.

```
v = L / t
Power = F * v          (AGM 206: linear power = force x speed)
```

v = ________________ in/s Power = ________________ hp

11. Operator input linear power

At the same rate, using the hand force and the arc distance the hand travels, determine the linear power exerted by the operator. Neglect the return stroke.

```
d = 2 * pi * b * theta / 360          [theta from Table 2, in degrees]
Power = F_hand * d * n
```

d = ________________ in Power = ________________ hp

> **Expect a small mismatch.** The arc is slightly longer than the straight-line displacement the lever ratio implies, by a factor of theta/sin(theta) with theta in radians, typically 6 to 7 percent. Your input power will come out a few percent above questions 8 and 10. This is a geometry artifact of the arc formula, not a loss and not an error. Say so in question 13.

12. Reservoir area and drop in fluid level

(a) Determine the cross-sectional area of the donut-shaped reservoir, then find the drop in fluid level when the lifting cylinder is fully extended. **(b)** Using the height of the annular oil space, determine the maximum reservoir capacity and show whether the reservoir can supply a full extension. Report the margin.

```
A_res    = (pi/4) * (D_o^2 - D_i^2)
V_lift   = A_r * L
drop     = V_lift / A_res
capacity = A_res * H
margin   = capacity / V_lift        must exceed 1.0
```

A_res = ________________ in^2 drop = ________________ in

capacity = ________________ in^3 margin = ________________ : 1

> **Two checks on your own numbers.** The drop cannot exceed the oil column height, or the pump would draw air before full extension. And the annulus is interrupted by the pump cylinder boss and the cast passages, so a clean donut overstates the true capacity; treat your answer as an upper bound and say so.

13. Comparison of input, hydraulic, and output power

Compare your answers from questions 8, 10, and 11. Has the jack magnified power? By what factor is force multiplied, and by what factor is speed reduced? Show that the total force factor is the product of the hydraulic ratio and the lever ratio, and account for the mismatch noted in question 11.

```
force factor  = W / F_hand      and      (A_r / A_p) * (b / a)
speed factor  = hand speed / ram speed
```


9 Report

Submit Tables 1 through 3, all calculations with substitutions shown, your corrected flow-path sketch from Part C and a closing paragraph answering this: which of the eight core principles could you have confirmed by measurement in this lab, and which had to be taken on the authority of the equations because the jacks contain no fluid? Be specific about what an oil-filled jack would have let you measure that this one did not.

10 Citation

Cite this activity and its accompanying simulation as:

1. Koc, A. B. (2026). *Interactive Hydraulic Bottle Jack Cross Section and Laboratory Materials* (Version 1.2.0) [Computer software and instructional materials]. Agricultural Sciences Department, Agricultural Mechanization and Business Program, Clemson University. Zenodo. `https://doi.org/10.5281/zenodo.22181633`